Chapter 10. Attention and Transformers

Lecture 11/2: attention-based captioning, spatial and channel attention, inductive bias, scaled dot-product, cross- and self-attention, positional encoding, masking, multi-head attention, Transformer encoder, layer normalization

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1 Attention-Based Image Captioning

Motivation. Each generated word re-weights the image regions it describes.

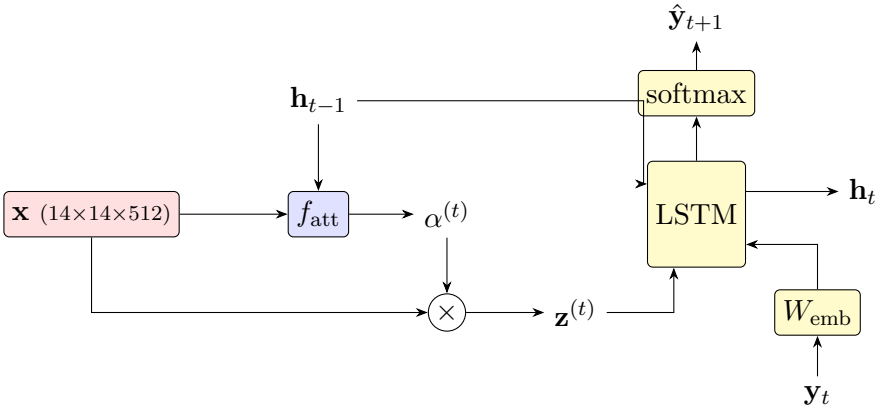
Keep VGG's conv5 map $\mathbf{x} \in \mathbb{R}^{14\times14\times512}$ ($224 \times 224 \times 3$ input): $L = 196$ vectors $\mathbf{x}_{i,j} \in \mathbb{R}^{512}$. Each word t has an **attention map** $\alpha^{(t)} \in \mathbb{R}^{14\times14}$, nonnegative with $\sum_{i,j} \alpha^{(t)}_{i,j} = 1$ ($\alpha^{(1)}$ for "a", $\alpha^{(2)}$ for "bird", ...), summarizing the image as $\mathbf{z}^{(t)} = \sum_{i,j} \alpha^{(t)}_{i,j}\mathbf{x}_{i,j} \in \mathbb{R}^{512}$.

Definition 10.1: attention-based captioning (Show, Attend and Tell)

At step t, with previous hidden state $\mathbf{h}_{t-1}$ and current input word $\mathbf{y}_t$:

$$
\begin{aligned}
e^{(t)} &= f_{\text{att}}(\mathbf{x}, \mathbf{h}_{t-1}) && \text{(alignment scores, one per location)},\\
\alpha^{(t)}_{i,j} &= \frac{\exp(e^{(t)}_{i,j})}{\sum_{i',j'} \exp(e^{(t)}_{i',j'})} && \text{(attention weights)},\\
\mathbf{z}^{(t)} &= \sum_{i,j} \alpha^{(t)}_{i,j}\mathbf{x}_{i,j} && \text{(context vector)},\\
\mathbf{h}_t &= \text{LSTM}(\mathbf{h}_{t-1}, W_{\text{emb}}\mathbf{y}_t, \mathbf{z}^{(t)}) && \text{(modified LSTM)},\\
\hat{\mathbf{y}}_{t+1} &\propto \exp\left(W^o(W_{\text{emb}}\mathbf{y}_t + W^h\mathbf{h}_t + W^z\mathbf{z}^{(t)})\right) && \text{(next-word distribution)}.
\end{aligned}
$$

The context vector enters every LSTM gate; f_{att} is a small MLP; the cell repeats once per word.



$(\mathbf{x}, \mathbf{h}_{t-1}) \mapsto \alpha^{(t)}$, *and* $\mathbf{z}^{(t)}$ *enters the LSTM.*

$\mathbf{z}^{(t)}$ is a weighted average; uniform weights $1/L$ give the naive model with average pooling.

Soft versus hard attention. Soft attention (above) is differentiable and trained by backpropagation. **Hard attention** samples $s_t \sim \text{Categorical}(\alpha^{(t)})$, $\mathbf{z}^{(t)} = \mathbf{x}_{s_t}$ (one spot per word, vs. soft spreading over bird and water); with no gradient through sampling it needs score-function (REINFORCE) estimators.

2 Spatial and Channel Attention

Weights on the axes of $U \in \mathbb{R}^{H\times W\times C}$:

- **Spatial attention** weights the $H \times W$ locations (Section 1).
- **Channel attention** weights the C channels.

Definition 10.2: squeeze-and-excitation block (SENet)

Given $U = [\mathbf{u}_1, \ldots, \mathbf{u}_C]$ with $\mathbf{u}_c \in \mathbb{R}^{H \times W}$:

$$
\begin{aligned}
s_c &= \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i,j) && \text{(squeeze: global average pooling)}, \\
\mathbf{e} &= \sigma\big(W_2 \, \mathrm{ReLU}(W_1 \mathbf{s})\big) && \text{(excitation)}, \\
\tilde{\mathbf{u}}_c &= e_c \, \mathbf{u}_c && \text{(scale)},
\end{aligned}
$$

with $W_1 \in \mathbb{R}^{(C/r) \times C}$ and $W_2 \in \mathbb{R}^{C \times (C/r)}$.

Sigmoid weights $e_c \in (0, 1)$ need not sum to 1. The block adds $2C^2/r$ weights (r e.g. 16), inside residual (SE-ResNet) or mobile (ANTNets) blocks.

3 Inductive Bias

Motivation. Each architecture builds in assumptions; attention has the fewest.

Four families:

- MLP: linear layers and nonlinear functions.
- CNN: convolution, pooling, nonlinearities and an MLP head.
- RNN: MLPs for the recurrent update, plus gating functions.
- Attention and Transformers: dot-product attention plus MLPs.

Data and models can be mixed (1D CNN for sequences, RNN or MLP on pixels).

Definition 10.3: inductive bias

The **inductive bias** of a learner is the set of assumptions it uses to predict outputs for inputs it has not observed.

Mostly **locality** and **invariance**:

- Convolution: 2D grid, translation equivariance (Chapter 7).
- RNN: 1D chain, first-order Markov ($\mathbf{h}_t$ sees the past only via $\mathbf{h}_{t-1}$).

Efficient, but not transferable across modalities; one model for images, point clouds, audio and language needs few structural assumptions.

4 The General Attention Layer

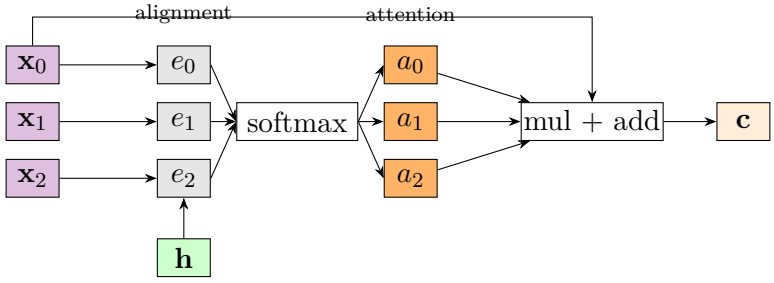
4.1 One query

Definition 10.4: attention with one query

Inputs: vectors $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^D$ (stacked as $\mathbf{x} \in \mathbb{R}^{N \times D}$) and a query $\mathbf{h} \in \mathbb{R}^D$.

$$\begin{aligned} \text{alignment:} \quad & e_i = f_{\text{att}}(\mathbf{h}, \mathbf{x}_i) = \frac{\mathbf{h} \cdot \mathbf{x}_i}{\sqrt{D}}, \\ \text{attention:} \quad & \mathbf{a} = \text{softmax}(\mathbf{e}), \\ \text{output:} \quad & \mathbf{c} = \sum_{i=1}^{N} a_i \mathbf{x}_i \in \mathbb{R}^D \quad \text{(context vector)}. \end{aligned}$$

An $H \times W \times D$ map gives $N = HW$ vectors. Reordering them leaves the sum $\mathbf{c}$ unchanged (**permutation invariance**).



One query $\mathbf{h}$*: scores, softmax, weighted sum.*

4.2 Why divide by $\sqrt{D}$

For independent entries of $\mathbf{h}, \mathbf{x}$ with mean 0, variance 1, $z = \mathbf{h} \cdot \mathbf{x} = \sum_{m=1}^{D} h_m x_m$ has $\mathbb{E}z = 0$, $\text{Var}\, z = \sum_m \text{Var}(h_m x_m) = D$. Unscaled logits of sd $\sqrt{D}$ make the softmax nearly one-hot, with almost no gradient through the weights; dividing by $\sqrt{D}$ restores unit variance.

4.3 Many queries, keys and values

Definition 10.5: multiple queries

Queries $\mathbf{q}_1, \dots, \mathbf{q}_M \in \mathbb{R}^D$ ($\mathbf{q} \in \mathbb{R}^{M \times D}$), one context vector each:

$$e_{i,j} = \frac{\mathbf{q}_j \cdot \mathbf{x}_i}{\sqrt{D}}, \qquad \mathbf{a}_{\cdot,j} = \text{softmax}(\mathbf{e}_{\cdot,j}), \qquad \mathbf{y}_j = \sum_i a_{i,j} \mathbf{x}_i.$$

$\mathbf{x}$ plays two roles (alignment, output); a separate linear layer for each adds expressiveness and changes dimensions.

Definition 10.6: cross-attention layer

With learnable $W_k \in \mathbb{R}^{D \times D_k}$ and $W_v \in \mathbb{R}^{D \times D_v}$ and queries $\mathbf{q} \in \mathbb{R}^{M \times D_k}$:

$$\mathbf{k} = \mathbf{x} W_k, \quad \mathbf{v} = \mathbf{x} W_v, \quad e_{i,j} = \frac{\mathbf{q}_j \cdot \mathbf{k}_i}{\sqrt{D_k}}, \quad \mathbf{a}_{\cdot,j} = \text{softmax}(\mathbf{e}_{\cdot,j}), \quad \mathbf{y}_j = \sum_i a_{i,j} \mathbf{v}_i \in \mathbb{R}^{D_v}.$$

Definition 10.7: self-attention layer

With $\mathbf{q} = \mathbf{x}W_q$, $W_q \in \mathbb{R}^{D\times D_k}$, the layer maps N inputs to N outputs (**self-attention**); rows as tokens,

$$\mathrm{SA}(\mathbf{x}) = \mathrm{Attn}(\mathbf{x}W_q, \mathbf{x}W_k, \mathbf{x}W_v) = \mathrm{softmax}\Big(\frac{\mathbf{x}W_q(\mathbf{x}W_k)^\top}{\sqrt{D_k}}\Big)\mathbf{x}W_v,$$

softmax by rows: the Transformer's **scaled dot-product attention** $\mathrm{Attn}(Q,K,V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})V$.

$A = \mathrm{softmax}(QK^\top/\sqrt{d_k}) \in \mathbb{R}^{n\times n}$ is row-stochastic, so each output averages all values: a contextual encoding with whole-sequence receptive field (**feature aggregation**).

The four steps. $\mathbf{x} \in \mathbb{R}^{n\times d}$, $W_q, W_k \in \mathbb{R}^{d\times d_k}$, $W_v \in \mathbb{R}^{d\times d_v}$; $Q = \mathbf{x}W_q$ (**query**, "the question"), $K = \mathbf{x}W_k$ (**key**, "the index"), $V = \mathbf{x}W_v$ (**value**, "the information"):

(i) $A = QK^\top \in \mathbb{R}^{n\times n}$, $A_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j$: how much token i uses token j;

(ii) $A' = QK^\top/\sqrt{d_k}$: $\mathrm{Var}\sum_{m=1}^{d_k} q_m k_m = d_k$ as above;

(iii) $\alpha_{ij} = \mathrm{softmax}(A')_{ij}$, row by row;

(iv) $\mathrm{output}_i = \sum_j \alpha_{ij}\mathbf{v}_j$.

Cross versus self. Cross-attention takes queries from elsewhere (a decoder), keys and values from X; self-attention takes all three from X. Captioning (Section 1) is cross-attention (query: sentence state; keys, values: image features). In Definition 10.4 keys and values coincide.

5 Permutation Symmetry

For a permutation matrix P, cross-attention with fixed queries is **permutation invariant** (keys and values reorder together), and self-attention is **permutation equivariant**, $\mathrm{SA}(P\mathbf{x}) = P\ \mathrm{SA}(\mathbf{x})$, since $\mathbf{q}_i \cdot \mathbf{k}_j$ depends only on the two tokens. "Dog bites man" and "man bites dog" agree up to reordering: order must be supplied separately.

6 Positional Encoding

Motivation. Attach a position vector to each token, since self-attention cannot see positions.

$\mathbf{p}_j = \mathrm{pos}(j) \in \mathbb{R}^d$ is concatenated (slides) or added, $\mathbf{x}_j + \mathbf{p}_j$ (Transformer). Options:

1. **Learn a lookup table**: $T \times d$ parameters for $t \in [0, T)$.
2. **Design a fixed function**.

Intuition: in a binary counter each bit is periodic in the position (periods $2, 4, \ldots$).

Definition 10.8: sinusoidal positional encoding

For even d and $k = 0, 1, \ldots, \frac{d}{2} - 1$, let $\omega_k = 1/10000^{2k/d}$ and

$$\mathrm{PE}(t)_{2k} = \sin(\omega_k t), \qquad \mathrm{PE}(t)_{2k+1} = \cos(\omega_k t).$$

Equivalently, $\mathrm{PE}(t) = \big(\sin\omega_0 t, \cos\omega_0 t,\ \ldots,\ \sin\omega_{d/2-1}t, \cos\omega_{d/2-1}t\big)$.

Unlike a table it is defined for every t; frequencies fall geometrically from 1 to about $1/10000$, fast to slow like counter bits.

Example 10.9: $d = 4$ (lecture example)

$\omega_0 = 1$, $\omega_1 = 1/10000^{2/4} = 0.01$: $\mathrm{PE}(t) = (\sin t, \cos t, \sin 0.01t, \cos 0.01t)$ and

$$\mathrm{PE}(0) = (0, 1, 0, 1), \qquad \mathrm{PE}(1) \approx (0.841, 0.540, 0.010, 1.000),$$
$$\mathrm{PE}(2) \approx (0.909, -0.416, 0.020, 1.000).$$

With $\mathbf{z}_j = \mathbf{x}_j + \mathbf{p}_j$ and $\mathbf{x}_{\text{dog}} = (1, 0, 2, 0)$,

$$\mathbf{x}_{\text{dog}} + \mathrm{PE}(0) = (1, 1, 2, 1), \qquad \mathbf{x}_{\text{dog}} + \mathrm{PE}(2) \approx (1.909, -0.416, 2.020, 1.000),$$

so "dog" at positions 0 and 2 differ, and "dog bites man" is no longer a reordering of "man bites dog".

7 Masked Self-Attention

Generation must not see future words. Training on "man plays guitar", the Transformer predicts all next tokens in parallel:

position	input	target
0	`<SOS>`	man
1	man	plays
2	plays	guitar
3	guitar	`<EOS>`

Position 1 could copy its target "plays" from position 2, so query j may attend only to $0, \ldots, j$: future scores are set to $-\infty$ ($e^{-\infty} = 0$).

Definition 10.10: masked (causal) self-attention

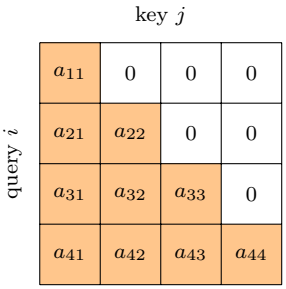
With $\mathbf{q}_i, \mathbf{k}_j, \mathbf{v}_j$ as in Definition 10.7, query i uses the scores

$$e_{i,j} = \begin{cases} \mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{D_k}, & j \le i, \\ -\infty, & j > i, \end{cases} \qquad \mathbf{y}_i = \sum_j a_{i,j} \mathbf{v}_j, \quad \mathbf{a}_{i,\cdot} = \mathrm{softmax}(\mathbf{e}_{i,\cdot}).$$

The slides put the key index first, $e_{i,j} = \mathbf{q}_j \cdot \mathbf{k}_i / \sqrt{D_k}$ ($i \le j$), $-\infty$ ($i > j$):

$$E = \begin{pmatrix} e_{0,0} & e_{0,1} & e_{0,2} \\ -\infty & e_{1,1} & e_{1,2} \\ -\infty & -\infty & e_{2,2} \end{pmatrix}$$

(column j = query j), the transpose of the pattern below. Rows still sum to 1; position i uses only tokens $1, \ldots, i$.



Causal attention: each query sees itself and earlier tokens.

8 Multi-Head Attention

Definition 10.11: multi-head attention

With h heads, each with its own projections $W_q^{(r)}, W_k^{(r)} \in \mathbb{R}^{d\times d_k}$ and $W_v^{(r)} \in \mathbb{R}^{d\times d_v}$,

$$\text{head}_r = \text{Attn}\left(QW_q^{(r)}, KW_k^{(r)}, VW_v^{(r)}\right), \qquad \text{MHA}(Q,K,V) = [\text{head}_1, \ldots, \text{head}_h]\, W_o,$$

with $W_o \in \mathbb{R}^{hd_v\times d}$. Usually $d_k = d_v = d/h$.

Each head (MatMul, Scale, optional Mask, SoftMax, MatMul) can track a different relation (previous word, subject).

9 The Transformer Encoder

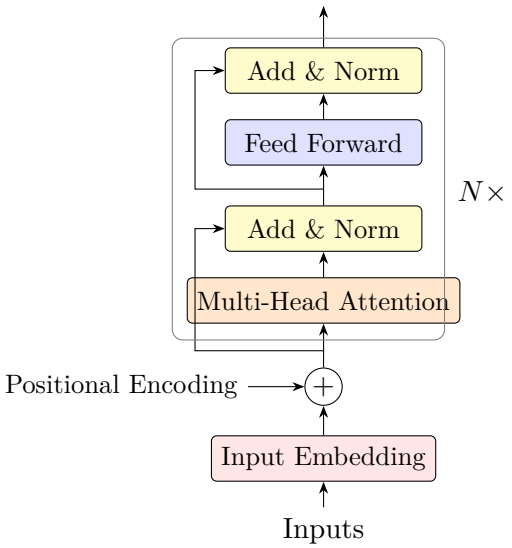
Motivation. Self-attention and position-wise MLPs, each with skip connection and layer norm (Vaswani et al., 2017).

Definition 10.12: Transformer encoder

1. Tokenize and embed: $\mathbf{x} \in \mathbb{R}^{n\times d}$.
2. Add positions: $X_0 = \mathbf{x} + \text{PE}$.
3. Each of N layers:

$$Z = \text{LN}\left(X + \text{MHA}(X,X,X)\right), \qquad X' = \text{LN}\left(Z + \text{FFN}(Z)\right),$$

   $\text{FFN}(\mathbf{z}) = W_2\,\text{ReLU}(W_1\mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2$ per token.
4. Layers are chained.



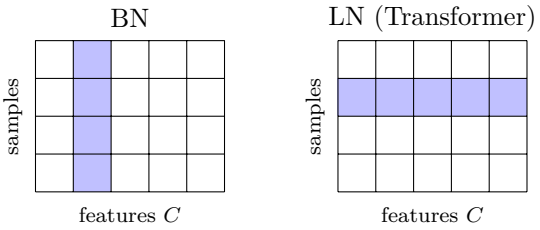
One encoder layer (boxed), repeated N times; side arrows are skips.

Positions enter only through PE. Cost is dominated by $QK^\top \in \mathbb{R}^{n\times n}$: $O(n^2 d)$ time, $O(n^2)$ memory.

10 Layer Normalization

BN (Chapter 6) uses batch statistics per channel; LN normalizes each sample by itself.

- **BN**: per channel, over batch N and positions H, W.
- **LN in CNNs**: per sample, over C, H, W.
- **LN in Transformers**: per sample and token, over the C features.



Shaded cells share one mean and variance.

Definition 10.13: layer normalization

For $\mathbf{x} \in \mathbb{R}^d$, let $\mu = \frac{1}{d}\sum_k x_k$ and $\sigma^2 = \frac{1}{d}\sum_k (x_k - \mu)^2$. Then

$$\hat{\mathbf{x}} = \frac{\mathbf{x} - \mu \mathbf{1}}{\sqrt{\sigma^2 + \epsilon}}, \qquad \mathrm{LN}(\mathbf{x}) = \boldsymbol{\gamma} \odot \hat{\mathbf{x}} + \boldsymbol{\beta},$$

with learnable $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^d$ and a small $\epsilon > 0$.

11 Why Transformers

Built for translation, the Transformer scales well with data and underlies BERT, GPT-2, Megatron-LM, T5, Turing-NLG (up to 17 billion parameters by early 2020). Its weak inductive bias costs on small data and pays off on large data.